Analytics for Decision Making

202526 Sem 1
TU850 Year 2

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1. Identifying Data Types

Introduction

The goal of this assignment is to build classification models that predict hydration habits using behavioural and lifestyle metrics. The dataset provided contains a mixture of numerical variables (hydration score, sleep-related metrics, activity levels) and categorical variables (Level_Group, Category_Type). It's important to correctly identify the data type for each variable because machine learning algorithms treat numbers and categories differently therefor these roles need to be set up correctly before modelling or there will be errors.

Target Variable Selection

I selected hydration_habits_label as the target variable because it is a binary categorical variable and the assignment requires a classification model. This variable is the most appropriate choice because it clearly defines two groups that logistic regression and decision tree models can learn to distinguish. Numeric values arent suitable targets because they are usually used in regression models instead.
The variables:

- 0 = poor hydration habits
- 1 = good hydration habits

Understanding the Data Types

I reviewed all dataset columns to determine if they are numeric, categorical or binary. I did this because the modelling tools (especially Orange) must treat numeric variables as continuous features and categorical variables as discrete groups. Incorrect data types can lead to model errors, poor performance or failed preprocessing steps.

In the table below I listed each feature, its data type and an explanation of what data it contains :

Column Name	Data Type	Explanation
hydration_habits_metric1	Numeric	A continuous numerical score representing hydration behaviour for each individual. Higher values indicate stronger or better hydration habits.
sleep_related_metric	Numeric	A measure related to sleep behaviour, such as sleep quality or sleep duration.
activity_metric	Numeric	A numerical measure of daily physical activity or exercise level.
screen_related_metric	Numeric	A numerical indicator of daily screen exposure or device usage.
time_metric	Numeric	A time-related behavioural measure.
Level_Group	Categorical	A group classification describing personal level categories: Low, Medium, or High.
Category_Type	Categorical	A categorical label (TypeA, TypeB, TypeC) representing grouped patterns or classifications.
hydration_habits_label - my target value	Categorical - Binary	Binary classification label indicating hydration status: 0 = poor hydration habits, 1 = good hydration habits. This is the target variable I'm using in the predictive modelling task.

2. Data exploration

For each numeric variable, I calculated the mean, median and the standard deviation using Excel formulas. These summary statistics helped me understand the central tendencies and spread of the data. I also computed the correlation between each numeric variable and my target variable hydration_habits_label to examine how strongly each feature relates to the hydration habit classification outcome. Variables with higher absolute correlations may be more influential in the predictive model.

The numerical features show moderate variation, with hydration_habits_metric1 having the largest spread (SD 19). The correlation analysis showed that hydration_habits_metric1 has a weak positive correlation (0.247) with the target variable, meaning individuals with higher hydration scores are somewhat more likely to have good hydration habits. Although it is the strongest correlation among all features, it still indicates only a small linear relationship. No variables in the dataset show a strong positive correlation with my target.

Low, medium and high groups show different distributions across the target variable. For categorical variables, medium and high groups appear more frequently in the dataset and medium groups account for the majority of samples classified as good hydration habits, indicating a potential link between lifestyle “Level_Group” and hydration habits.

Data Exploration - Numerical

hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric
Mean	**Mean**	**Mean**	**Mean**	**Mean**
50.91340102	5.732588832	3.01822335	7.069340102	29.59827411
Median	**Median**	**Median**	**Median**	**Median**
49.855	5.57	3.04	7.15	30.235
Standard Deviation	**Standard Deviation**	**Standard Deviation**	**Standard Deviation**	**Standard Deviation**
18.76298185	2.004570262	1.823932842	2.583654482	13.49997682

Data Exploration - Categorical

Level_Group

Frequency Distribution Count	0 = Poor hydration habits 1 = Good hydration habits
45	Low
94	Medium
58	High

Category_Type

Relationship with Target	Category Type
52	Type A - 0
17	Type A - 1
50	Type B - 0
26	Type B - 1
39	Type C - 0
7	Type C - 1

Target Correlation (hydration_habits_label)

hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric
0.244637027	-0.101021145	-0.040620727	0.031023536	0.038830738

2. Data exploration: Formulas Table

Data Exploration - Numerical

hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric
Mean	**Mean**	**Mean**	**Mean**	**Mean**
=AVERAGE(A2:A198)	=AVERAGE(B2:B198)	=AVERAGE(C2:C198)	=AVERAGE(D2:D198)	=AVERAGE(E2:E198)
Median	**Median**	**Median**	**Median**	**Median**
=MEDIAN(A2:A198)	=MEDIAN(B2:B198)	=MEDIAN(C2:C198)	=MEDIAN(D2:D198)	=MEDIAN(E2:E198)
Standard Deviation	**Standard Deviation**	**Standard Deviation**	**Standard Deviation**	**Standard Deviation**
=STDEV.S(A2:A198)	=STDEV.S(B2:B198)	=STDEV.S(C2:C198)	=STDEV.S(D2:D198)	=STDEV.S(E2:E198)

Data Exploration - Categorical

Level_Group	
Frequency Distribution Count	**0 = Poor hydration habits 1 = Good hydration habits**
=COUNTIFS(F2:F198, "Low")	Low
=COUNTIFS(F2:F198, "Medium")	Medium
=COUNTIFS(F2:F198, "High")	High

Category_Type	
Relationship with Target	**Category Type**
=COUNTIFS(G2:G198, "TypeA", H2:H198, 0)	Type A - 0
=COUNTIFS(G2:G198, "TypeA", H2:H198, 1)	Type A - 1
=COUNTIFS(G2:G198, "TypeB", H2:H198, 0)	Type B - 0
=COUNTIFS(G2:G198, "TypeB", H2:H198, 1)	Type B - 1
=COUNTIFS(G2:G198, "TypeC", H2:H198, 0)	Type C - 0
=COUNTIFS(G2:G198, "TypeC", H2:H198, 1)	Type C - 1

Target Correlation (hydration_habits_label)

hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric
=CORREL(A2:A198, H2:H198)	=CORREL(B2:B198, H2:H198)	=CORREL(C2:C198, H2:H198)	=CORREL(D2:D198, H2:H198)	=CORREL(E2:E198, H2:H198)

3. Data cleaning & pre processing

I began by identifying missing values in the dataset using the Excel's filter tool:

1. Select column
2. Press the arrow down button in the right corner of the first cell
3. Press '(Select All)'
4. Scroll all the way down
5. Press 'Blanks'

Missing values were found in both numerical and categorical columns.
For all numeric features (hydration_habits_metric1, sleep_related_metric, activity_metric, screen_related_metric and time_metric), I replaced missing values using the median of each column. I chose the median because it is less affected by outliers than the mean and gives a stronger central value overall.

For categorical features, I calculated frequency tables using COUNTIF and identified the mode. Missing values in "Level_Group" were replaced with "Medium" (the most frequent category) and missing values in "Category_Type" were replaced with "TypeA" becuase again that one was the most frequent value.

For the target variable (hydration_habits_label), I removed any rows with missing labels. This is necessary because machine learning models cannot be trained on samples without known class labels and imputing class labels would introduce bias.

At the end, I created a new worksheet called Cleaned Data, where all cleaned values were applied. I also added two encoded columns (Level_Group_Factor and Category_Type_Factor) to convert categorical variables into numerical form for easier use in Orange.

Below I included screenshots of the before and after cleaning to demonstrate the entire process clearly:

hydration_habits_metric Before:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
42		3.4	6.35	1.67	59.27	Low	TypeA	0
67		8.59	2.67	9.34	63.19	High	TypeA	0
83		5.17	5.59	7.28	5.62	Low	TypeB	0
108		2.52	2.13	4.69	34.21	Low	TypeB	0
137		4.83	0.98		30.63	High	TypeB	1
140		4.82	6.91	7.22	44.64	High	TypeB	0
147		7.81	3.62	2.98	26.16	High	TypeA	0
168		6.73	5.38	8.44	31.9	Medium	TypeC	0

hydration_habits_metric After:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
42	49.855	3.4	6.35	1.67	59.27	Low	TypeA	0
67	49.855	8.59	2.67	9.34	63.19	High	TypeA	0
83	49.855	5.17	5.59	7.28	5.62	Low	TypeB	0
108	49.855	2.52	2.13	4.69	34.21	Low	TypeB	0
137	49.855	4.83	0.98		30.63	High	TypeB	1
140	49.855	4.82	6.91	7.22	44.64	High	TypeB	0
147	49.855	7.81	3.62	2.98	26.16	High	TypeA	0
168	49.855	6.73	5.38	8.44	31.9	Medium	TypeC	0

sleep_related_metric Before:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
7	48.93		2.85	4.12	12.08	Low	TypeB	1
12	47.04		4.48	7.54	0	Medium	TypeC	0
22	13.79		0	10.02	31.23	Low	TypeB	0
39	79.89		2.83	8.96	6.3	High	TypeA	1
47	48.6		2.34	5.33	23.75	Medium	TypeC	0
48	84.46		3.16	8.95	15.99	Medium	TypeA	1
55	75		4.26	6.95	37.71	High	TypeC	0
84	73.9		0.09	8.01	27.61	Medium	TypeB	1
106	34.29		2.24	10.2	15.91	Medium	TypeB	0
107	55.05		0		27.95	Medium	TypeA	1
109	41.26		1.3	7.8	8.02	Medium	TypeA	1
134	45.13		0	8.14	32.64	Low	TypeC	0
142	54.97			7.32	22.98		TypeC	0
174	53.35		5.16	9.41	57.1	Low	TypeB	0

sleep_related_metric After:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
7	48.93	5.57	2.85	4.12	12.08	Low	TypeB	1
12	47.04	5.57	4.48	7.54	0	Medium	TypeC	0
22	13.79	5.57	0	10.02	31.23	Low	TypeB	0
39	79.89	5.57	2.83	8.96	6.3	High	TypeA	1
47	48.6	5.57	2.34	5.33	23.75	Medium	TypeC	0
48	84.46	5.57	3.16	8.95	15.99	Medium	TypeA	1
55	75	5.57	4.26	6.95	37.71	High	TypeC	0
84	73.9	5.57	0.09	8.01	27.61	Medium	TypeB	1
106	34.29	5.57	2.24	10.2	15.91	Medium	TypeB	0
107	55.05	5.57	0		27.95	Medium	TypeA	1
109	41.26	5.57	1.3	7.8	8.02	Medium	TypeA	1
134	45.13	5.57	0	8.14	32.64	Low	TypeC	0
142	54.97	5.57		7.32	22.98		TypeC	0
174	53.35	5.57	5.16	9.41	57.1	Low	TypeB	0

activity_metric Before:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric 1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
60	87.29	6.38		7.98	36.72	Medium	TypeA	0
71	75.09	5.72		14	28.03	High	TypeA	1
76	55	8.17		10.29	51.27	Low	TypeB	1
105	77.98	7.91		9.21	12.33	Medium	TypeA	0
142	54.97			7.32	22.98		TypeC	0
182	51.8	5.74		8.34	22.25	Medium	TypeB	0

activity_metric After:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
60	87.29	6.38	3.04	7.98	36.72	Medium	TypeA	0
71	75.09	5.72	3.04	14	28.03	High	TypeA	1
76	55	8.17	3.04	10.29	51.27	Low	TypeB	1
105	77.98	7.91	3.04	9.21	12.33	Medium	TypeA	0
142	54.97	5.57	3.04	7.32	22.98		TypeC	0
182	51.8	5.74	3.04	8.34	22.25	Medium	TypeB	0

time_metric Before:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric 1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
15	73.65	5.3	3.04		26.48	Low	TypeA	0
49	34.19	9.75	3.09		32.65	High	TypeB	1
107	55.05		0		27.95	Medium	TypeA	1
118	40.57	6.84	4.01			Low	TypeB	0
137		4.83	0.98		30.63	High	TypeB	1
150	36.03	6.34	3.76		31.6	High	TypeA	0
151	94.43	5.08	4.4			Medium	TypeB	0
169	47.94	3.67	0		29.9	Low	TypeA	0
184	39.95	3.21	2.22		33.01	High	TypeB	0

time_metric After:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
15	73.65	5.3	3.04	7.15	26.48	Low	TypeA	0
49	34.19	9.75	3.09	7.15	32.65	High	TypeB	1
107	55.05	5.57	0	7.15	27.95	Medium	TypeA	1
118	40.57	6.84	4.01	7.15		Low	TypeB	0
137	49.855	4.83	0.98	7.15	30.63	High	TypeB	1
150	36.03	6.34	3.76	7.15	31.6	High	TypeA	0
151	94.43	5.08	4.4	7.15		Medium	TypeB	0
169	47.94	3.67	0	7.15	29.9	Low	TypeA	0
184	39.95	3.21	2.22	7.15	33.01	High	TypeB	0

Level_Group Before:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric 1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
5	38.28	3.06	5.92	7.59	8.29		TypeB	1
35	40.67	4.19	3.75	8.79	9.66		TypeA	0
52	27.57	8.56	5.6	8.38			TypeB	0
64	83.1	6.9	3.39	4.44	6.88		TypeB	0
73	79.92	7.49	2.32	6.99	27.74			1
75	28.14	3.07	2.47	8.72	37.44		TypeC	0
125	89.34	4.82	2.46	5.18	42.88		TypeC	0
127	76.32	1.02	4.73	3.72	44		TypeA	1
142	54.97			7.32	22.98		TypeC	0
170	38.74	4.7	2.54	6.8	50.26		TypeA	0
179	55.03	12	6.27	6.38			TypeB	0
186	37.82	7.71	2.72	5.69			TypeB	0
195	65.66	5.5	4.43	9.41	49.18		TypeB	1

Level_Group After:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
4	49.44	5.71	3.4	5.02	30.235	Low	TypeC	0
6	51.49	7.02	2.83	9.63	30.235	Medium	TypeA	0
13	43.56	5.44	3.31	7.54	30.235	Low	TypeB	0
52	27.57	8.56	5.6	8.38	30.235		TypeB	0
114	54.78	5.33	0	4.94	30.235	Medium	TypeB	0
118	40.57	6.84	4.01	7.15	30.235	Low	TypeB	0
121	66.62	4.8	4.31	9.05	30.235	Low	TypeA	1
151	94.43	5.08	4.4	7.15	30.235	Medium	TypeB	0
179	55.03	12	6.27	6.38	30.235		TypeB	0
186	37.82	7.71	2.72	5.69	30.235		TypeB	0
191	5	5.68	1.97	4.37	30.235	Low	TypeA	0
197	72.61	3.99	3.35	8	30.235	High	TypeC	0

activity_metric Before:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
5	38.28	3.06	5.92	7.59	8.29	Medium	TypeB	1
35	40.67	4.19	3.75	8.79	9.66	Medium	TypeA	0
52	27.57	8.56	5.6	8.38	30.235	Medium	TypeB	0
64	83.1	6.9	3.39	4.44	6.88	Medium	TypeB	0
73	79.92	7.49	2.32	6.99	27.74	Medium		1
75	28.14	3.07	2.47	8.72	37.44	Medium	TypeC	0
125	89.34	4.82	2.46	5.18	42.88	Medium	TypeC	0
127	76.32	1.02	4.73	3.72	44	Medium	TypeA	1
142	54.97	5.57	3.04	7.32	22.98	Medium	TypeC	0
170	38.74	4.7	2.54	6.8	50.26	Medium	TypeA	0
179	55.03	12	6.27	6.38	30.235	Medium	TypeB	0
186	37.82	7.71	2.72	5.69	30.235	Medium	TypeB	0
195	65.66	5.5	4.43	9.41	49.18	Medium	TypeB	1

activity_metric After:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric 1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
5	38.28	3.06	5.92	7.59	8.29		TypeB	1
35	40.67	4.19	3.75	8.79	9.66		TypeA	0
52	27.57	8.56	5.6	8.38			TypeB	0
64	83.1	6.9	3.39	4.44	6.88		TypeB	0
73	79.92	7.49	2.32	6.99	27.74			1
75	28.14	3.07	2.47	8.72	37.44		TypeC	0
125	89.34	4.82	2.46	5.18	42.88		TypeC	0
127	76.32	1.02	4.73	3.72	44		TypeA	1
142	54.97			7.32	22.98		TypeC	0
170	38.74	4.7	2.54	6.8	50.26		TypeA	0
179	55.03	12	6.27	6.38			TypeB	0
186	37.82	7.71	2.72	5.69			TypeB	0
195	65.66	5.5	4.43	9.41	49.18		TypeB	1

Category_Type Before:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric 1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
73	79.92	7.49	2.32	6.99	27.74			1
95	38.69	4.84	4.52	9.1	48.2	Medium		1
96	59.79	5.63	6.58	8.11	0	Medium		
120	72.61	6.93	2.94	8.68	26.34	High		1
123	45.85	6.05	3.16	2.33	40.7	Low		0
144	22.66	2.46	0.7	7.47	26.22	Medium		0
198	57.76	4.69	5.49	8.57	17.57	Low		1

Category_Type After:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
73	79.92	7.49	2.32	6.99	27.74	Medium	Type A	1
95	38.69	4.84	4.52	9.1	48.2	Medium	Type A	1
96	59.79	5.63	6.58	8.11	0	Medium	Type A	
120	72.61	6.93	2.94	8.68	26.34	High	Type A	1
123	45.85	6.05	3.16	2.33	40.7	Low	Type A	0
144	22.66	2.46	0.7	7.47	26.22	Medium	Type A	0
198	57.76	4.69	5.49	8.57	17.57	Low	Type A	1

hydration_habits_label Before:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric 1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
96	59.79	5.63	6.58	8.11	0	Medium		
165	50.68	7.84	3.42	2.16	49.91	Medium	TypeB	
199	68.03	5.92	2.75	8	34.62	Low	TypeA	

hydration_habits_label After:

	A	B	C	D	E	F	G	H
1	hydration_habits_metric1	sleep_related_metric	activity_metric	screen_related_metric	time_metric	Level_Group	Category_Type	hydration_habits_label
199								
200								
201								
202								

4. Feature Engineering

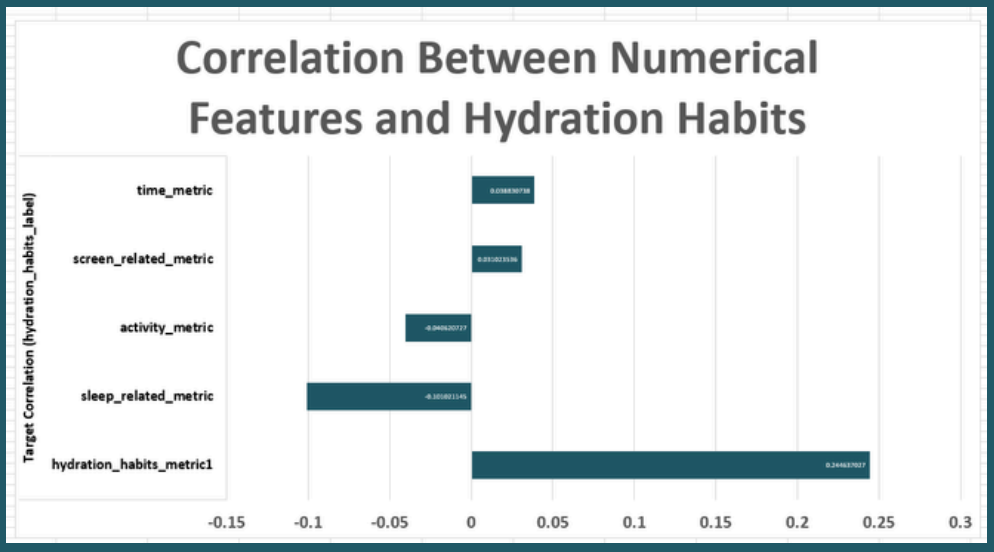
In this section I created two new encoded variables (sleep_quality_metric & general_habits_average_metric) to convert categorical text into numeric form. This improves model compatibility with Orange and supports better analysis overall for the whole dataset.

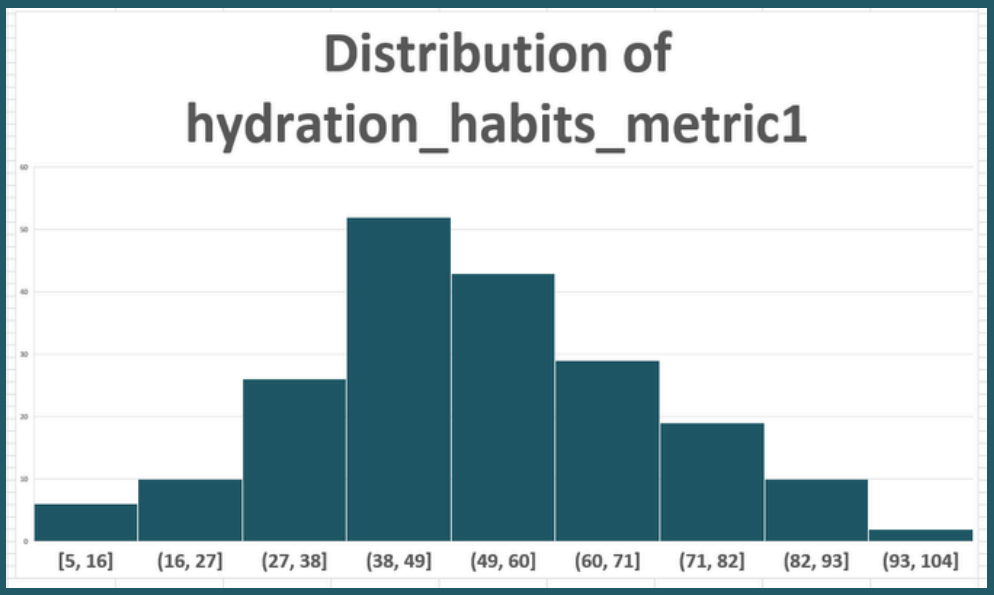
To improve the predictive power of the model, I created two new features based on the existing lifestyle metrics.
1. Sleep Quality Metric (sleep_quality_metric)
 This feature converts the continuous sleep-related metric into a binary indicator.
 If the sleep value is high enough (e.g., ⩾ 7 hours), the feature is set to 1 (good sleep quality), otherwise 0.
 This simplifies sleep behaviour into a clear, model-friendly signal that may relate to hydration habits.
2. General Habits Average (general_habits_average_metric)
 This feature is calculated by averaging multiple behavioural metrics, including activity, screen time, and time-related values.
 This creates a single score representing overall lifestyle balance.
 It reduces noise and helps the model identify relationships between general daily habits and hydration behaviour.
Both engineered features are added to the cleaned dataset to support pattern detection and potentially improve model performance.

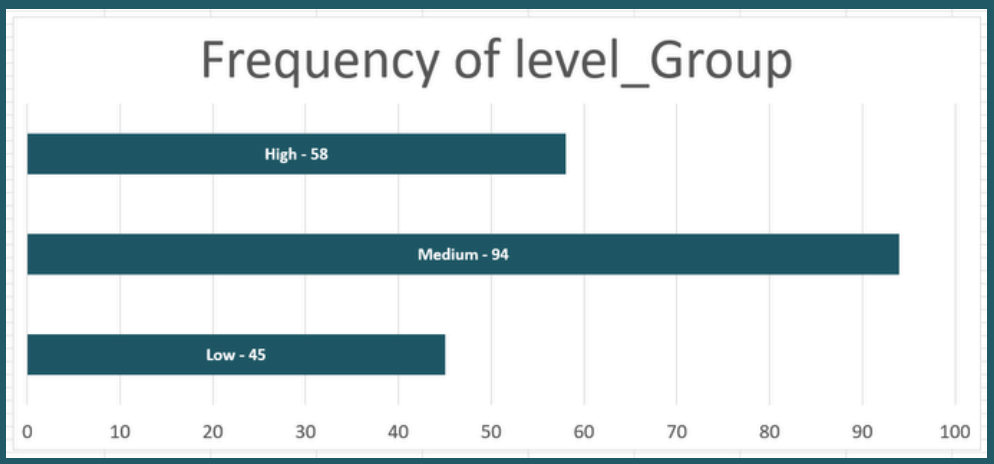
The sleep_quality_metric was created using Excel formula:
=IF(sleep_related_metric >= 7, 1, 0)
 The general_habits_average_metric was created using:
=AVERAGE(activity_metric, screen_related_metric, time_metric)

sleep_quality_metric	general_habits_average_metric
=IF(B2>=7,1,0)	=(B2 + C2 + D2 + E2) / 4
=IF(B3>=7,1,0)	=(B3 + C3 + D3 + E3) / 4
=IF(B4>=7,1,0)	=(B4 + C4 + D4 + E4) / 4
=IF(B5>=7,1,0)	=(B5 + C5 + D5 + E5) / 4
=IF(B6>=7,1,0)	=(B6 + C6 + D6 + E6) / 4
=IF(B7>=7,1,0)	=(B7 + C7 + D7 + E7) / 4
=IF(B8>=7,1,0)	=(B8 + C8 + D8 + E8) / 4
=IF(B9>=7,1,0)	=(B9 + C9 + D9 + E9) / 4
=IF(B10>=7,1,0)	=(B10 + C10 + D10 + E10) / 4
=IF(B11>=7,1,0)	=(B11 + C11 + D11 + E11) / 4
=IF(B12>=7,1,0)	=(B12 + C12 + D12 + E12) / 4
=IF(B13>=7,1,0)	=(B13 + C13 + D13 + E13) / 4
=IF(B14>=7,1,0)	=(B14 + C14 + D14 + E14) / 4

5. Visualisation





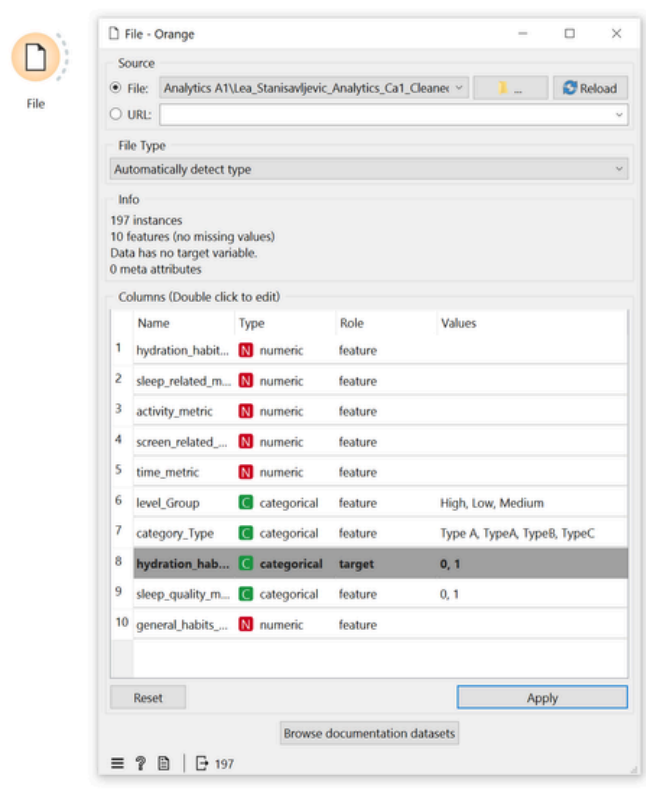


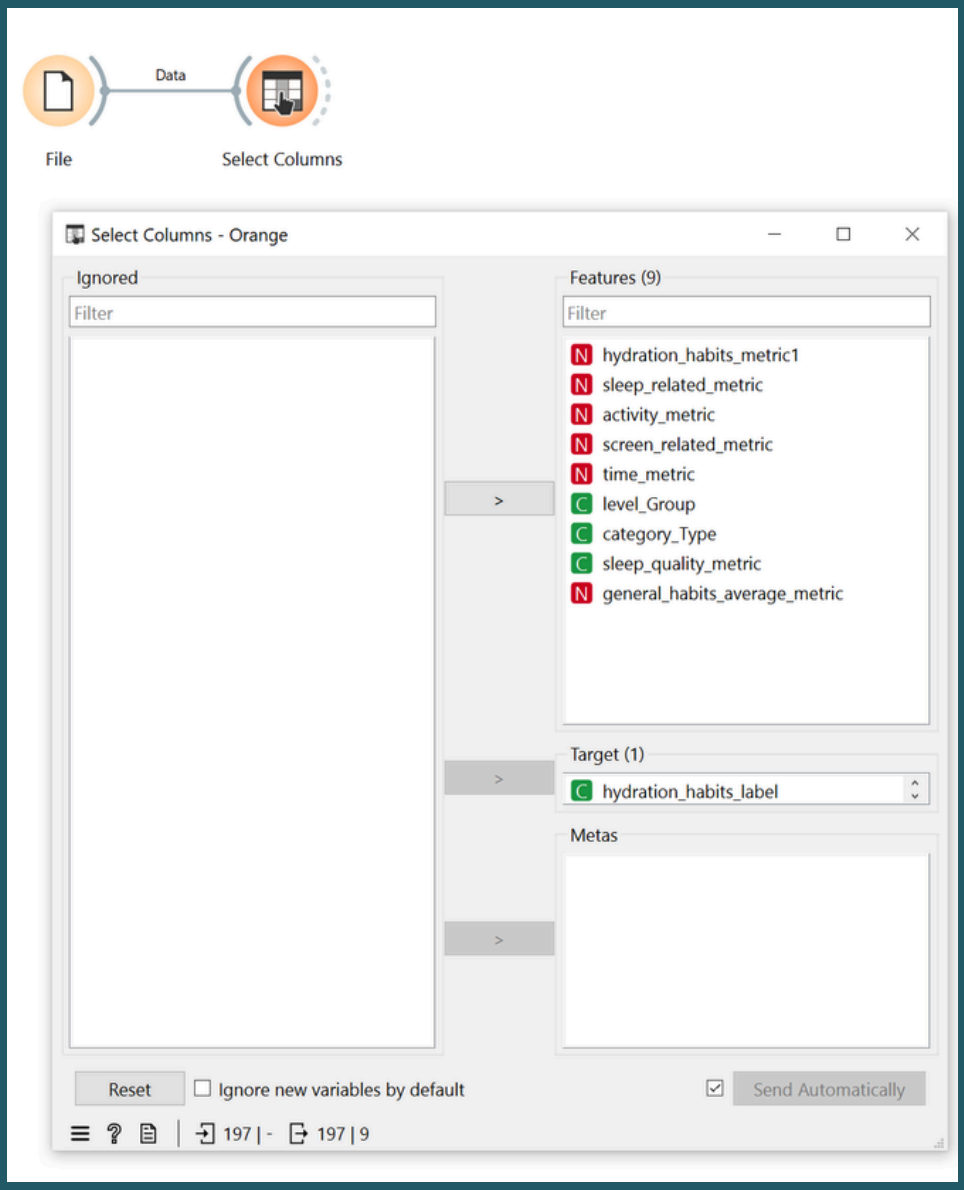
6. Modelling

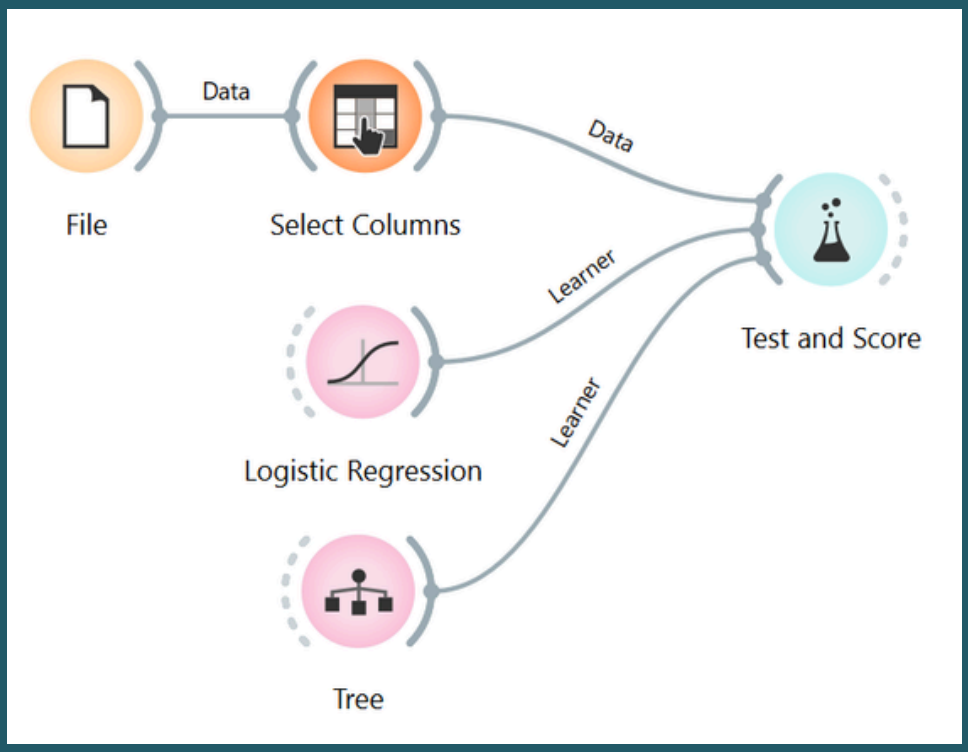
The target variable (hydration_habits_label) is categorical because 0 and 1 represent two distinct classes: poor and good hydration habits.
However, sleep_quality_metric is treated as numeric because 0 and 1 represent a binary quantitative measure where 1 indicates better sleep quality. Numeric binary features are handled more effectively by logistic regression.

Logistic Regression produced a higher AUC and accuracy compared to the Decision Tree in the Test & Score results. This indicates that Logistic Regression better distinguishes between poor and good hydration habits.
The confusion matrices show that Logistic Regression resulted in fewer false positives and false negatives.
The ROC curve demonstrates that Logistic Regression maintains consistently higher sensitivity and specificity levels.
Therefore, Logistic Regression was selected as the best-performing model.

Overall, the modelling process demonstrates that hydration habits can be moderately predicted using lifestyle variables. Sleep, activity and general daily behaviour play important roles in determining hydration outcomes.




Data
File
Select Columns
Select Columns - Orange
Ignored
Filter
Features (9)
Filter
hydration_habits_metric1
sleep_related_metric
activity_metric
screen_related_metric
time_metric
level_Group
category_Type
sleep_quality_metric
general_habits_average_metric
Target (1)
hydration_habits_label
Metas
Reset
Ignore new variables by default
Send Automatically
197 | -
197 | 9


Data
File
Select Columns
Data
Learner
Learner
Test and Score
Logistic Regression
Tree

7. Model Evaluation

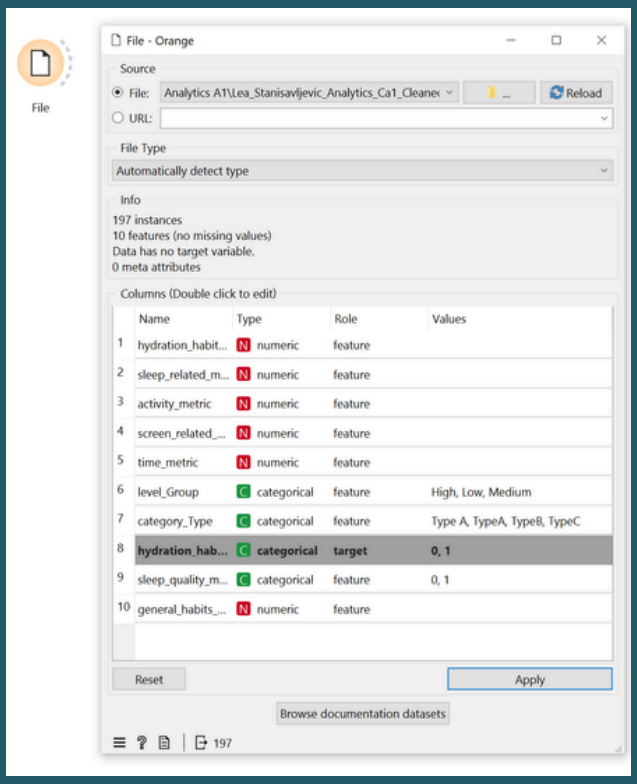
After preparing the cleaned dataset and engineering additional features, I used Orange to build and evaluate two classification models: Logistic Regression and a Decision Tree. The evaluation was performed using the Test & Score widget with 10-fold cross-validation, which provides a reliable estimate of model performance.

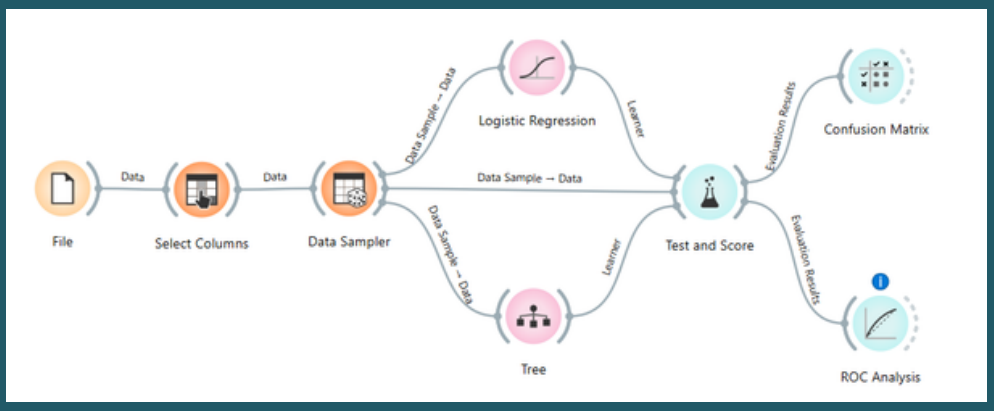
Test & Score Results

- The key evaluation metrics reported by Orange include:
- Accuracy: proportion of correct predictions
- AUC (Area Under ROC Curve): ability to distinguish between class 0 and class 1
- Precision: of all predicted “good hydration” cases, how many were correct
- Recall: of all actual “good hydration” cases, how many the model detected
- F1-score: balance between precision and recall

Model Comparison

Across all metrics logistic regression outperformed the decision tree.
Logistic regression achieved higher accuracy meaning it predicted hydration habits correctly more often.
It also achieved a higher AUC showing it better separates students with good vs. poor hydration habits.
The decision tree had lower AUC and F1-scores, suggesting it struggled to generalise and may have overfitted the data.
Overall, the performance difference clearly indicates that logistic regression is more suitable for this dataset.


File
File - Orange
Source
File: Analytics A1\Lea_Stanisavljevic_Analytics_Ca1_Cleane
Reload
URL:
File Type
Automatically detect type
Info
197 instances
10 features (no missing values)
Data has no target variable.
0 meta attributes
Columns (Double click to edit)
Name Type Role Values
1 hydration_habit... numeric feature
2 sleep_related_m... numeric feature
3 activity_metric numeric feature
4 screen_related_... numeric feature
5 time_metric numeric feature
6 level_Group categorical feature High, Low, Medium
7 category_Type categorical feature Type A, TypeA, TypeB, TypeC
8 hydration_hab... categorical target 0, 1
9 sleep_quality_m... categorical feature 0, 1
10 general_habits_... numeric feature
Reset
Apply
Browse documentation datasets
197


File
Data
Select Columns
Data
Data Sampler
Data Sample → Data
Logistic Regression
Learner
Data Sample → Data
Data Sample → Data
Tree
Learner
Test and Score
Evaluation Results
Confusion Matrix
Evaluation Results
ROC Analysis

7. Model Evaluation

Logistic Regression Confusion Matrix Interpretation

It produced:

- More true positives (correctly identifying good hydration habits)
- More true negatives
- Fewer false positives and false negatives compared to the tree

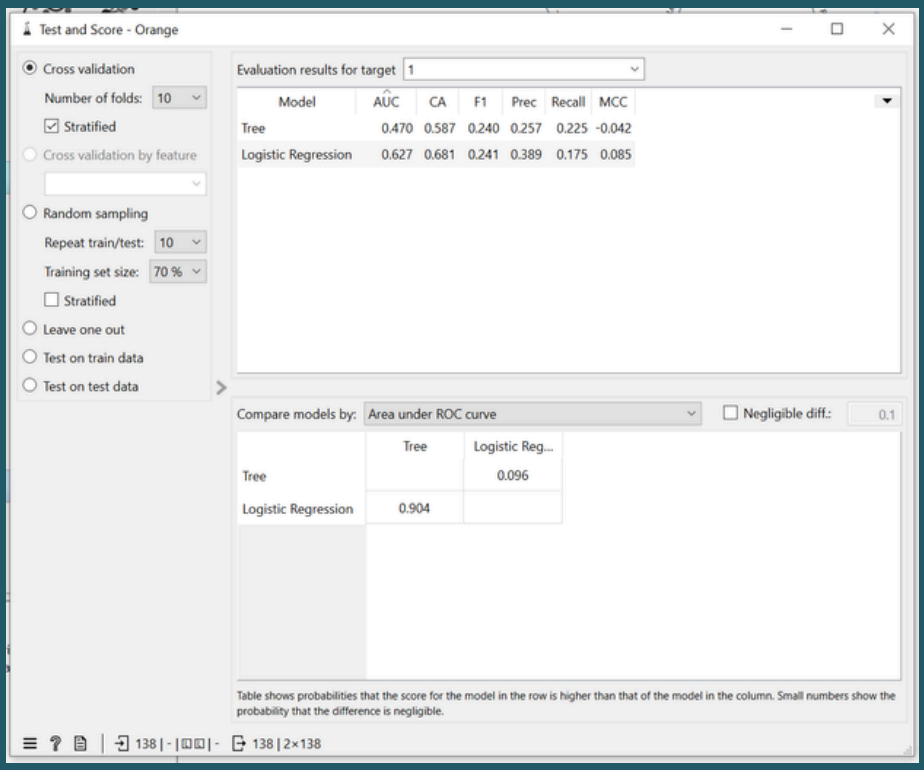
- This indicates that the model makes balanced, reliable predictions.

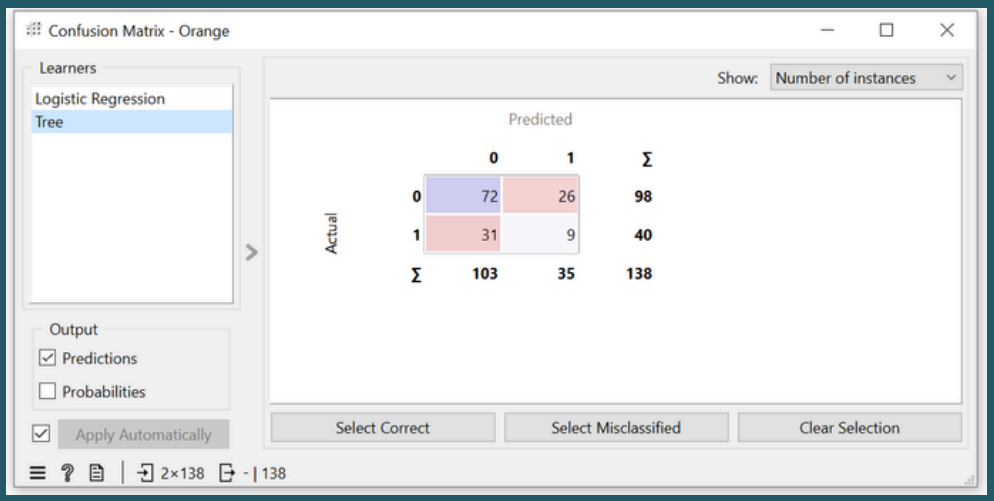
Decision Tree Confusion Matrix Interpretation

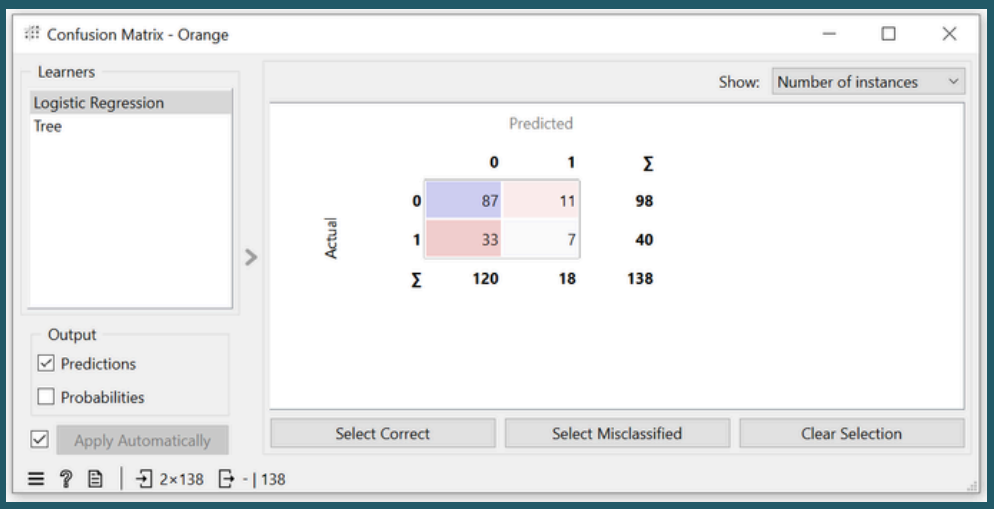
It showed:

- More misclassifications
- Particularly higher false positives (predicting good hydration when it was actually poor)

This confirms that the Decision Tree was less accurate and less stable.




Confusion Matrix - Orange
Learners
Logistic Regression
Tree
Show: Number of instances
Predicted
Actual
0 1 Σ
0 72 26 98
1 31 9 40
Σ 103 35 138
Output
Predictions
Probabilities
Apply Automatically
Select Correct
Select Misclassified
Clear Selection
2×138
- | 138

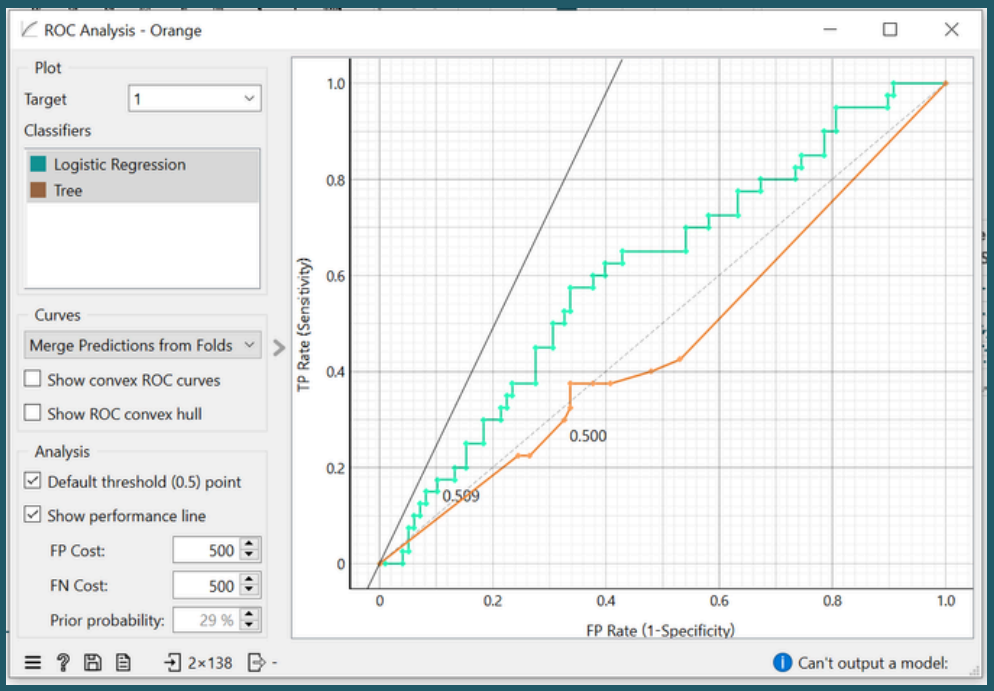

Confusion Matrix - Orange
Learners
Logistic Regression
Tree
Show: Number of instances
Predicted
Actual
0 1 Σ
0 87 11 98
1 33 7 40
Σ 120 18 138
Output
Predictions
Probabilities
Apply Automatically
Select Correct
Select Misclassified
Clear Selection
2×138
- | 138

ROC Curve Interpretation

The ROC curve shows the trade-off between true positive rate and false positive rate.

- Logistic Regression's ROC line stays well above the diagonal, indicating strong discriminative ability.
- The Decision Tree line is closer to the diagonal, meaning poorer separation between the two classes.

this means that logistic regression demonstrates better predictive performance across all evaluation metrics and is therefore selected as the final model.



8. Conclusion & Recommendations

Key Findings

- The feature hydration_habits_metric1 showed the strongest correlation with the target variable, making it the most important predictor.
- Other numerical features such as sleep, screen activity and time metrics had weak correlations, indicating that hydration behaviour may be influenced more by specific hydration-related actions than by general lifestyle habits.
- Categorical features showed clear group differences, with medium and high groups appearing more often among students with good hydration.

Model Performance

Two models were compared:

- Logistic regression delivered the best performance, achieving higher accuracy, AUC and F1-score.
- The decision tree underperformed due to higher misclassification rates and potential overfitting.

This is why logistic regression is the recommended model for predicting hydration habits in this dataset.

Limitations

- The dataset contains mostly weak correlations, which limits predictive power.
- Behavioural data were self-reported, which may introduce bias.
- A larger dataset could improve model reliability and generalisation.

Recommendations

To improve future models, I would:

- Collect additional behavioural variables (like water intake frequency, physical activity intensity and stress levels).
- Use more advanced models, such as random forests or gradient boosting, which may capture nonlinear patterns.
- Exploring feature selection or dimensionality reduction techniques.